Scalar Implicatures and Implicit Metalinguistic Negation*

Jae-Il Yeom
(Hongik University)

Yeom, Jae-Il. 2008. **Scalar implicatures and implicit metalinguistic negation.** *Korean Journal of English Language and Linguistics* 8-4, 631-662. The Neo-Gricean account of scalar implicatures is globalistic in that a scalar implicature of an utterance is derived by negating a stronger alternative as a whole. The account is problematic in dealing with scalar implicatures in complex sentences. Recently various localistic approaches have been proposed to solve the problems. In this paper, I show that scalar implicatures cannot be accounted for by strictly globalistic or strictly localistic approaches. Sentences with the same syntactic and semantic structures yield local scalar implicatures at some times and global scalar implicatures at others. I propose a more flexible approach in which the negation operator which negates a stronger alternative is assumed to be metalinguistic negation and that a set of scalar implicatures is obtained depending on which part is inappropriate to assert in the speaker's information state. More informative scalar implicatures are preferred unless they are incompatible with the context.

Key Words: scalar implicature, global vs. local, metalinguistic negation, Neo-Gricean, disjunction

1. Introduction

The Neo-Gricean account of scalar implicatures is based on Grice's maxim of quantity, and the idea has been developed by

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Horn (1972, 1989), Gazdar (1979), Atlas and Levinson (1981), Harnish (1976), Hirschberg (1985), etc. Recently their ideas have been criticized as having some problems in predicting scalar implicatures in complex sentences. It has been pointed out that the main problems come from the globalistic position of the Neo-Gricean analysis. However, localistic approaches are not solutions. In this paper, I will claim that the basic mechanism must be globalistic, but that actual scalar implicatures are derived from a set of possible contexts the speaker implicates. I will start with the introduction of the Neo-Gricean account of scalar implicatures.

Horn (1972) observes that a scalar implicature arises from a set of alternatives which forms a scale, where one member is stronger or weaker than another. The scale is known to every conversational participant in a speech context, and selecting one alternative makes the hearer suppose that the speaker is not in a position to utter a sentence with a stronger alternative.[2] In (1a), the use of or invokes the set of scalar alternatives [or, and]. And the hearer supposes that the speaker is not in a position to assert (1b).

(1) a. John is laughing or crying.
b. John is both laughing and crying.
c. It is not the case that (= NOT) John is both laughing and crying.

The reason could be that the speaker does not have evidence for doing so or that the speaker has evidence that the stronger

[2]A scale arises only because one form is stronger than another. But the strength should be defined pragmatically as well as semantically. For this reason, the notion of informativeness may be preferred. In addition, scales are formed contextually as well as conventionally. Fauconnier (1975) and Hirschberg (1985) showed that some arbitrary scales can be established in speech contexts.

proposition does not hold. This may lead to the implicature that the speaker does not know that John is doing both. But more preferably, assuming that the speaker knows whether or not an alternative sentence for each scalar alternative is true, the hearer supposes that the speaker implicates (1c), if there is no reason to doubt the assumption. So the utterance of (1a) is understood as meaning that John is laughing or crying, but not doing both.

A scalar implicature arises even when a scalar term is used in the scope of the negation operator, as illustrated below.

(2) a. John read many books.
    b. John did not read many books.

If (2a) is uttered, the use of many invokes the set of scalar alternatives [no, some, many, (most), all/every] implicates that John did not read all books. On the other hand, if (2b) is uttered, it is implicated that John read some books. The same reasoning as in (1) works in this case too. If the speaker were informed that John did not read some (preferably, any) books, he or she would say so because the speaker could be more cooperative by saying so. Since the speaker didn't say so, we make the inference that it is not the case that John read no books, namely, that John read some books. The existence of the negation operator reverses the direction of the implicature: in ordinary cases, all is stronger than many, but in D(downward-)E(entailing) contexts some is stronger than them. (Fauconnier 1975)

Levinson (2000) made these observations: in DE contexts, scalar implicatures are induced by inverse scales. This is quite natural in the Neo-Gricean analysis, because in this analysis strength is compared with respect to whole utterances. Regardless of whether the negation or other DE operator is involved in a sentence, a scalar implicature is derived from a stronger alternative to the assertion, and it depends on the monotonicity of the operators involved in the sentence whether the replacement of a scalar term with another gives us a stronger utterance. In this sense, the Neo-Gricean account of scalar implicatures can be called a *globalistic* analysis.

It has been pointed out recently that a globalistic account is problematic in dealing with utterances with scalar terms embedded in some operators. The most typical example is a disjunctive sentence.

(3) Mary walked or read some poems in the book.

(4) a. Mary walked or read all poems in the book.

b. NOT(Mary walked or read all poems in the book)
(= Mary didn't walk nor read all poems in the book.)

c. Mary walked or read some, but not all, poems in the book.

In this example, there are two scalar terms involved. By the use of *or*, the speaker implicates that Mary did not both walk and read some poems in the book. But here we are more concerned with the use of *some* in the second disjunct.[3] If (3) is uttered, we might expect to get a scalar implicature like (4b) by negating the corresponding stronger alternative (4a). However, this is incompatible with mentioning the first disjunct in (3). If the speaker knows that John did not walk, he or she would have said simply that Mary read some poems in the book. The

[3]The existential quantifier some may not give rise to a scalar implicature. Consider the following example.

i. John read some books on the table.

If some books on the table are all there are in the current context, the use of some does not yield the implicature of not all. Milsark (1974), Diesing (1992) and Horn (1997) distinguished weak "sum" from strong "some": only the latter participates in a quantitative scale.

actual understanding of the assertion is something like (4c). This is not what we can get from a globalistic account.

Some attempts have been made to avoid this problem by Chierchia (2004), Sauerland (2004), etc. Chierchia (2004) proposed a localistic analysis. Sauerland (2004) basically takes the globalistic position, but it is localistic in dealing with disjunctive sentences by assuming a different alternative set of the disjunction operator. In the next section, I will discuss problems with the Neo-Gricean account in detail, with more focus on Sauerland (2004). In Section 3, I will review localistic accounts, focusing on Chierchia (2004). In section 4, I will propose my own analysis which is devoid of any problems which previous analyses show.

2. Problems with Neo-Gricean Accounts of Implicatures

When a scalar term gives rise to a scalar implicature in the utterance of a simplex sentence, can it also be a scalar implicature in the utterance of a complex sentence in which the simplex sentence is embedded? Neo-Griceans did not pay much attention to this question, and gave the wrong generalizations. Gazdar (1979) claimed that scalar implicatures are blocked in embedded contexts. Hirschberg (1985/1991) claimed that scalar implicatures are blocked by overt negation alone. As we have seen in (2), a scalar implicature can arise in a context in the scope of the negation operator. This is what is expected by the Neo-Gricean account of scalar implicatures as a globalistic theory. A problem comes from cases where scalar implicatures seem to be derived locally. In this section, I will discuss this problem.

2.1 Disjunction

The problem of disjunction construction in the Neo-Gricean account, which is illustrated in (3-4), was first mentioned by McCawley (1993), and discussed in detail by Simons (1998), Chierchia (2004), and Fox (2006). To solve this problem, Sauerland (2004) proposed an idea in which each disjunct can yield a scalar implicature of the whole utterance. One crucial assumption in his analysis is that a disjunctive sentence 'S_1 or S_2' has four alternatives: {S_1 and S_2, S_1, S_2, S_1 or S_2}, not {S_1 and S_2, S_1 or S_2}. Each disjunct is taken to be a stronger alternative to the disjunctive sentence itself. And a stronger alternative to each disjunct, namely the conjunction sentence, is taken to be a stronger alternative to the whole sentence. This is illustrated in (5). The scalar term *or* implicates that Kai did not eat both. We are more concerned with the scalar implicature from *some*. Sauerland assumes that intuitively, the use of *some* implicates that Kai didn't have all of the peas last night.

(5) a. Kai had the broccoli or some of the peas last night.
b. Kai didn't have the broccoli and some of the peas last night.
c. Kai didn't have all of the peas last night.

To explain this intuition, he gets the following stronger alternatives to the sentence:

(6) a. Kai had some of the peas last night.
b. Kai had the broccoli last night.
c. Kai had the broccoli and some of the peas last night.
d. Kai had all of the peas last night.
e. Kai had the broccoli or all of the peas last night.
f. Kai had the broccoli and all of the peas last night.

Among these, (6a,b,e) do not survive because the negations of them will deny one of the two disjuncts.[4] The negation of (6c) gives rise to a scalar implicature from the use of *or*. And the negation of (6d) is taken to be a scalar implicature of the whole sentence from the use of *some*. This seems correct when the two disjuncts are not related to each other. Finally the negation of (6f) is entailed by the implicature from (6d), so it does not lead to any new implicature.

Sauerland predicts that scalar implicatures from each disjunct become scalar implicatures of whole sentences unless they are incompatible with the context. However, this is the case only when a disjunct is independent of the other. There are cases where an implicature from one disjunct should be accepted only when the other disjunct does not hold. Scalar implicatures we get in such cases must remain within the scope of the disjunct where they arise. Here are such examples.

(7) Kai had the broccoli or some of the peas last night. If he had the broccoli, he had more/all of the peas.

In (7), if the use of *some* generates the implicature that Kai did not have all of the peas, it will make the first disjunct false because the conditional entails that if Kai didn't have all of the peas, he would not have had broccoli. So the implicature should be canceled. However, we still think that the implicature from *some* holds within the second disjunct: the disjunctive sentence is

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[4]For every stronger alternative S′ for a sentence S, an implicature of the form 'the speaker doesn't know that S′ (=$\neg$K(S′)) is derived. Such implicatures are called primary scalar implicatures. Gazdar's clausal implicatures are included in the category of primary scalar implicatures. The stronger alternatives in (6) give rise to primary scalar implicatures. Each primary scalar implicature $\neg$K(S′) can be strengthened into an implicature of the form 'the speaker knows that NOT S′ (= K($\neg$S′)), if it is compatible with the primary scalar implicatures plus the assertion. Such implicatures are called secondary scalar implicatures. So-called scalar implicatures in other analyses correspond to secondary scalar implicatures in his analysis.

understood as meaning that Kai had the broccoli or some, though not all, of the peas last night. According to Sauerland (2004), if a potential implicature is incompatible with the context, it is simply canceled, not allowing it to be weakened. So he predicts no surviving scalar implicature from the first disjunct, contrary to what actually happens.[5] Sauerland's account is basically globalistic, but he approaches the problem of disjunctive sentences by dealing with each disjunct as if they were simplex root sentences. It resembles a localistic account in this respect. This is what causes problems here.

2.2 Scalar implicatures of existential quantifiers

When a scalar implicature is generated in the scope of an existential quantifier, it gives rise to a different set of problems. In this subsection, I will discuss a problem with existential quantifiers. The Neo-Gricean analysis predicts overly strong scalar implicatures. Consider the following example.

(8) a. Some students who watched TV or played games failed maths.
b. NOT[Some students who watched TV and played games failed maths]
c. Some students who NOT[watched TV and played

[5]Sauerland (2004) predicts that scalar terms in conditionals will behave the same way as those in disjunctions. So according to Sauerland's account we would get (iia) from (i) as a scalar implicature, just as we get (5c) from (5a).

i. If John finishes his work by 2 o'clock, he will get some of the apples.
ii. a. John will not get all of the apples.
b. If John finishes his work by 2 o'clock, he will get some, but not all, of the apples.

On the other hand, whether John gets some or all of the apples is a matter of concern in situations where John finishes his work by 2 o'clock. So the implicature should be something like (iib). (iia) is not likely to be a scalar implicature of the conditional as a whole.

games] failed maths.

In (8a), the use of *some* gives rise to a scalar implicature like 'Not all students who watched TV or played games last night failed maths', ignoring the effect of the use of *or*. We are more interested in the scalar implicature from *or*. The Neo-Gricean account predicts that by the use of *or* the speaker implicates (8b). If this implicature is not canceled, the utterance means that only students who did *one* of the two things failed maths. However, if doing one of the two things makes students fail maths, doing both will do so too. The speaker simply intends to relate each way of wasting time to failure in maths. With this intention considered, the implicature we get is (8c). However, this is not obtained in a globalistic account like the Neo-Gricean approach, since negation only applies to the whole utterance.

3. Localistic Approaches to Scalar Implicatures

To overcome the shortcomings of the Neo-Gricean account, various versions of localistic accounts have been proposed. In Relevance Theory, embedded scalar implicatures are introduced to LF structures as additional syntactic material and are regarded as constituting (part of) truth-conditions of utterances. Such accounts are found in Sperber and Wilson (1995) and Recanati (2003). There is another group of linguists who propose an operator which is projected in the syntactic structure. In that kind of approaches, the operator combines with a set of alternatives which is generated from a scalar term, and yields scalar implicatures. The operator is an implicit 'ONLY' in Krifka (1995) and Fox (2006), or an implicit negation which only applies to a stronger alternative in Chierchia (2004). In these accounts, scalar implicatures are derived in the process of compositional interpretation of sentences. In this paper, I will review Chierchia (2004), but the points I make can apply to other localistic approaches.

Chierchia (2004) assumes that if a scalar term is in a quantifier, it moves to a certain place when the quantifier is raised. In (9), for example, the quantifier *some of her students* is adjoined to the local VP. Calculation of a scalar implicature from *some* applies to the LF (9b) of the sentence (9a).

(9) a. Mary is either working at her paper or seeing some of her students.

b. [$_{IP}$ Mary$_i$ [[$_{VP}$ t$_i$ is working at her paper] or [$_{VP}$ some of her students$_j$ [$_{VP}$ t$_i$ seeing t$_j$]]]]

Scalar implicatures are calculated compositionally together with ordinary semantic interpretation. For each constituent, a set of alternative readings is derived together with the ordinary semantics, following Krifka (1995).

(9′) b. [$_{IP}$ Maryi [[$_{VP}$ t$_i$ is working at her paper] or
[$_{VP}$ some of her studentsj [$_{VP}$ t$_i$ seeing t$_j$] & NOT(all of her students$_j$ [$_{VP}$ t$_i$ seeing t$_j$])]]]

For this reason he does not need a separate projection mechanism for scalar implicatures. In (9′b), a scalar implicature is calculated by negating a stronger alternative locally when the VP with the quantifier adjoined is interpreted. So we get the meaning that Mary is working at her paper or seeing some, though not all, of her students. Scalar implicatures derived like this are called *direct scalar implicatures*. A localistic approach can account for cases where a scalar term is embedded in the scope of an existential quantifier, as in (8). Localistic approaches can account for problems with globalistic approaches, but it does not mean that localistic approaches are quite right.

According to Chierchia (2004), if compositional interpretation runs into a DE operator, a scalar implicature is blocked. In (10a), the scalar term *or* is used in the antecedent clause, and does not generate a scalar implicature. In (10b), on the other hand, the scalar term or is used in the consequent clause and generates a scalar implicature. This is because the antecedent clause of a conditional introduces a DE context, while the consequent clause does not.

(10) a. If Paul or Bill comes, Mary will be upset.
 b. If Paul comes, Mary or Sue will be upset.

Over the scope of a DE context, a new scalar implicature is added in non-locally from the interaction with the alternatives of scalar terms. But this time again, a scalar implicature is obtained by negating a new stronger alternative with the scale reversed. Chierchia illustrates it with the examples in (11). (11a) generates no scalar implicature, but (11b), in which the verb *doubt* introduces a DE context, generates the implicature that the speaker believes John drinks or smokes.

(11) a. John drinks and smokes.
 b. I doubt John drinks and smokes.

He assumes that *doubt* means 'not believe'. Replacement of *and* with or makes the whole sentence stronger, and the implicature is obtained by negating the whole alternative sentence. Scalar implicatures we get by this process are called *indirect scalar implicatures*.

In brief, Chierchia divides scalar implicatures into direct and indirect ones. A scalar term not embedded in a DE context generates a direct scalar implicature by local negation, but a scalar term embedded in a DE context generates an indirect scalar implicature by negating a stronger alternative of the constituent immediately dominating the DE operator when this has the widest scope among the DE operators.

His account makes correct predictions in calculating scalar implicatures in the disjunctive sentences and the sentences with epistemic operators which we have discussed. Considering indirect scalar implicatures, however, the account cannot be strictly local. In (12), for example, the operator introducing a DE context *no* has the whole sentence as its scope, so the SI Chierchia derives is the same as the one a globalistic approach predicts.

(12) No students who watched TV and played games last night have passed the exam.

When a scalar term is embedded in an existential quantifier, Chierchia (2004) points out that global negation leads to an overly strong scalar implicature, as in (8b). Since Chierchia also allows non-local negation in deriving indirect scalar implicatures, the same problem can occur when a scalar term is embedded in a DE operator. The same meaning as (8a) can be expressed with the negation having the widest scope, as in (13a).

(13) a. It is not the case that no students who watched TV or played games failed maths.
b. NOT[It is not the case that no students who watched TV and played games failed maths]
(= No students who watched TV and played games failed maths.)

In this case an implicature is predicted to be something like (13b), which is not what we want. Even in this case, the plausible implicature is obtained by local negation.

Second, Chierchia claimed that direct scalar implicatures are derived by local negation. However, this claim is not supported empirically, either. Since it is a contextual matter whether a scalar implicature is canceled or not, we can come up with an example in which a scalar implicature predicted by a globalistic account is not canceled.

(14) Some students who drank some of the beer were allowed to drive.

(15) a. NOT[some students who drank all of the beer were allowed to drive] (= No students who drank all of the beer were allowed to drive.)

b. Some students who NOT[drank all of the beer] were allowed to drive.

Neo-Griceans predict that the utterance of (14) implicates (15a). The SI is likely to survive. So the scalar implicature predicted by the Neo-Gricean account seems to be correct in this case. Chierchia predicts the implicature (15b). This is also plausible, but it is trivial, compared to (15a).

There are other examples in which global negation leads to more plausible scalar implicatures.

(16) a. Every student at MIT has read LGB or Syntactic Structures. (Sauerland 2004, 58)

b. NOT[Every student at MIT has read LGB and Syntactic Structures]

c. Every student at MIT NOT[has read LGB and Syntactic Structures]

(= No students at MIT have read LGB and Syntactic Structures.)

(17) a. Everyone must solve some of the problems.

b. NOT[everyone must solve all of the problems.

c. Everyone must NOT[solve all of the problems]
(= No one must solve all of the problems.)

(18) a. It's possible that Paul ate some of the eggs. (Hirschberg 1985, 75a)

b. NOT[It's possible that Paul ate all of the eggs]

c. It is possible that NOT[Paul ate all of the eggs]

In (16), students at MIT are likely to read as many syntax books as they can, and the two books are very important in studying the transformational syntax. So a student who reads one of the two is not likely to miss the second. So what we can infer from the statement is that not every student at MIT has read both, not that no students have read both. In (17), local negation leads to the implicature that no one must solve all of the problems, which is implausible. A more plausible one like (17b) is obtained by global negation. (18) is another example which shows that global negation leads to a more plausible scalar implicature. If someone says that Paul might have eaten some of the eggs, we normally understand it to implicate that Paul cannot have eaten all of the eggs, not that Paul might not have eaten all of the eggs.

So far I have pointed out that globalistic approaches like the Neo-Gricean account are problematic in dealing with disjunctive sentences, sentences with existential quantifiers, and sentences with propositional attitude verbs. The problems come from global negation in calculating scalar implicatures. Even Sauerland's (2004) account, which works like a localistic account in dealing with some disjunctive sentences, is globalistic in other cases and shows the same problems as other globalistic accounts. I have also discussed localistic approaches of scalar implicatures with focus on Chierchia (2004), and have shown that localistic approaches cannot be strictly local. In many cases, local negation does not give rise to correct scalar implicatures. Neither a strictly localistic nor a strictly globalistic account is a correct way of dealing with scalar implicatures. They argue for local negation in some linguistic contexts and global negation in other linguistic contexts. However, actual observations show that in the same linguistic contexts global negation is necessary for some cases and local negation is for other cases. This indicates that we cannot stick to one or the other between local and global negation, and that there should be flexibility to allow both possibilities in the same linguistic contexts. In the next section, I will propose an analysis of scalar implicatures that allows such flexibility.

4. Implicit Metalinguistic Negation for Implicating

4.1 Scalar implicatures as implicit metalinguistic negation

We have seen that neither a globalistic approach nor a localistic approach makes correct predictions. So we need to seek a less strict account which can predict local scalar implicatures in some cases and global implicatures in others. Pragmatic inferences are obtained from utterances, so the globalistic approach is a natural starting point, as the traditional account is. The source of the problems with previous globalistic approaches is that when they negate stronger alternatives to get scalar implicatures, they take the negation operator to be descriptive. Take a disjunction structure for example. In calculating scalar implicatures, it is assumed that a scalar term triggers a set of alternatives, making use of alternative semantics. Semantic interpretation deals with both ordinary semantics and sets of alternatives, as in Rooth (1985), Krifka (1995), etc., and the ordinary truth-conditional content of a sentence is considered against a relevant set of alternatives. For the use of *some* in

(19a), we can get the alternative set ALT_{some}((19a)), as given in (19b).

(19) a. John walked or read some of the short stories. (= A)
b. ALT_{some}(A) = {John walked or read some of the short stories,
John walked or read many of the short stories,
John walked or read all of the short stories}

The stronger alternatives are negated because it is assumed that the speaker is well informed and that he or she is not in a position to assert the stronger alternatives. The descriptive negation of one alternative leads to the negation of the first disjunct, but this should not be the case. All the alternatives allow for the possibility that the first disjunct holds, which means that the relevant context does not exclude that possibility. That is, the possibility that John walked should be taken for granted in the current context.

One way to keep the possibility of the first disjunct with global negation is to take the negation to be metalinguistic. One essential characteristic of metalinguistic negation is that the affirmative counterpart is echoic, as Carston (1996) claims. An echoic expression can include any expression which is not related to negation. Metalinguistic negation is used when the speaker wants to claim that it is inappropriate to assert the affirmative counterpart which is explicitly or implicitly invoked. Inappropriateness is not necessarily related to the notion of falsity. Consider the following examples.

(20) a. The king of France hasn't read some of his novels; there is no king of France.

b. Maggie's smart and attractive; not smart or attractive.

In (20a) metalinguistic negation is used to object to the use of the presupposition trigger, because the speaker believes that the presupposition does not hold. The negation operator indicates the inappropriateness of asserting the affirmative counterpart, rather than claim its falsity. However, this is possible when someone else makes the presupposition. In (20b) the speaker objects to the use of or because it conveys insufficient information, even though it is truth-conditionally true. If the affirmative counterpart is not invoked in the previous context, there is no reason to object to it.

Metalinguistic negation is also used when it is inappropriate to assert a sentence because only part of the sentence is not true. For example, (21) can be uttered only to negate the second disjunct. But we do this only when someone claims the affirmative counterpart.

(21) John didn't walk or read all of the short stories(, because it is not possible that he read all the short stories).

If we assume that the negation used in calculating scalar implicatures from disjunctive structures is metalinguistic, we do not have to specify which part is negated: the context restricted by an alternative set determines which part. In cases where scalar implicatures are calculated from conjunctive and disjunctive sentences, the alternative sets indicate that a disjunct where no scalar term is contained is taken for granted. So the negation goes to the disjunct in which a scalar term occurs. In (19) the negation operator applies to the stronger alternatives, but it only negates the second disjuncts of them.

I am proposing that the negation involved in calculating scalar implicatures is metalinguistic. Considering the fact that metalinguistic negation involves echoic expressions, it may be unusual to claim that a sentence not explicitly uttered is metalinguistically negated. But we can find such examples. Suppose that John was raising two mongooses, but John always used the plural form *mongeese*. Mary and Bill are friends of John's and they have met long time after they had seen John last.

(22) Mary: What is John doing these days?
Bill: He is still raising two mongeese. (He is not raising two mongooses.)

The correct alternative expression of "mongeese" is "mongooses" and what Bill implicates is that NOT(He is raising two mongooses). In this implicature, it is clear that the negation is metalinguistic.

When a scalar term is used and a scalar implicature is calculated, a stronger alternative is not actually mentioned. But when a scalar term is used in an utterance, the alternative set becomes salient to every conversational participant. In some sense, the speaker objects to using other alternatives to the scalar term actually used. There are two possible reasons for objecting to them. One is that some of them are not verified in the speaker's information state (violating the maxim of quality), and the other is that the others are not informative enough compared to the speaker's own information state (violating the maxim of quantity). These are illustrated in (23) and (24).

(23) A: Janet read some of Shakespeare's plays.
B: She read all.

(24) A: I like Sue. She's smart and attractive.

B: She's attractive.

In (23), B's utterance can be taken to be a correction of A's statement. In this case, B is implicating that Janet did not read some of the plays, where the negation is metalinguistic, just as in the following.

(25) A: Sue ate some of the cake.
B: She did not eat some of the cake. She ate all.

This is one of the typical examples of metalinguistic negation. In the example (23), the same metalinguistic negation seems to occur implicitly. The negation here does not negate the truth of A's statement, but it simply indicates that it is not appropriate, namely too weak, considering the actual state of affairs. This shows that metalinguistic negation plays a role in generating an implicature.

In (24), what B implicates is that Sue is not (both) smart and attractive.[6] By A's utterance, that Sue is smart and attractive is salient as the stronger alternative to B's utterance, and B's utterance is understood against the salient alternative, which is not verified in B's information state. In calculating the actual implicature, the implicit negation applies to the alternative as a whole only to negate the first conjunct. The negation operator is metalinguistic, even though it is not explicitly uttered. Combined

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[6]There are cases where and does not give rise to a scalar implicature as in (25).

i. A: He's a good hitter and doesn't use steroids.
B: He's (certainly) a good hitter.

In this example B's utterance does not implicate that he uses steroids. I suppose there is a way of deciding whether a scalar implicature arises: semantic/pragmatic relations between two conjuncts, intonation, the speaker's attitude toward the topic, etc.

with the assertion, it has the effect of negating only the first conjunct.[7]

4.2 Preference for stronger scalar implicatures

In the analysis proposed in this paper, the negation involved in calculating a scalar implicature is metalinguistic. It represents inappropriateness of some expression in the utterance. The next question to answer is what expression the negation is actually associated with in getting an inference. When an utterance is inappropriate to assert, there are various reasons for being so and the inappropriateness comes from anywhere in the sentence. However, cases involving scalar implicatures are restricted by sets of alternatives, so the target of association is always clear, and selecting one scalar term of limited strength implicates objection to the utterance of a stronger alternative. In conjunctive and disjunctive sentences, we have seen that the negation is associated with the conjunct/disjunct which contains a scalar term. This is rather a simple case because the use of a stronger alternative in one conjunct/disjunct does not affect the meaning of the other, unless the two conjuncts or disjuncts are related to each other.

Now consider cases where a scalar term is embedded in another operator whose interpretation can be affected by the use of an alternative of the scalar term.

(26) a. Every student read some of the stories.
b. ALT = {Every student read some of the stories,

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[7]If we assume that 'A or B' has four alternatives {'A and B', 'A', 'B', 'A or B'}, for a proposition q, there are an infinite number of stronger alternatives of the form p & q, p being any arbitrary proposition. This should not be the case. In our analysis of calculating scalar implicatures, such a problem does not arise. Only what is uttered previously by another speaker or implicitly invoked is relevant in calculating a scalar implicature. So there is no chance to allow for an infinite number of possible scalar implicatures.

Every student read many of the stories,
Every student read all of the stories}

c. NOT(every student read all of the stories) (NOT: descriptive negation)

d. It is inappropriate to assert that (= NOTM) every student read all of the stories. (NOTM: metalinguistic negation)

(27) a. Not every student read all of the stories.

b. Every student read not all of the stories.

In this example, there is no scope site whose interpretation is independent of the use of an alternative of the scalar term. If the negation operator used in a scalar implicature were descriptive, the implicature from the alternative *all* would be (26c) = (27a). On the other hand, if the operator is metalinguistic, the inappropriateness can be attributed, in principle, to any part including the stronger scalar term.[8]

When the speaker is not in a position to assert the stronger proposition with the use of *all*, the speaker can be in various information states. To see this, suppose the following four possible situations which represent the speaker's possible information states:

Situation 1: No students read all of the stories.

Situation 2: There are some students who read all of the stories, but the number of those students is not

[8]The stronger alternative with the stronger term may not be verified in the speaker's information state. It may, or may not, be falsified in the current information state. If a stronger alternative is assumed not to be verified in an information state, it leads to a 'don't-know' type implicature. As I claimed above, scalar implicatures of 'know-not' type of implicatures are stronger than implicatures of 'don't-know' type, which are obtained when the implicatures are falsified. We tend to get more information from the utterance by assuming that the stronger alternative is falsified, unless it is incompatible with the context.

significant.

Situation 3: There are students who read all of the stories, and the number of those students is significant.

Situation 4: Every student reads all of the stories.

Among the four, the sentence in question can be uttered in situations 1 and 2 without being criticized as violating the maxim of quantity. If (26a) were uttered only in Situation 1, the implicature would be only (27b). However, since it can be uttered in Situation 2 too, the scalar implicature can be (27a) or (27b).[9]

These observations show that possible information states in which (26a) can be uttered appropriately are associated with two different scopes of the negation operator with respect to the universal quantifier. Alternative scalar terms affect the meaning of the whole proposition, and the degree of the effect from an alternative scalar term is reflected in the scope of the negation operator. The role of metalinguistic negation in calculating scalar implicatures is that it just provides the minimal condition on the speaker's information state. We can come up with a set of possible information states which make a stronger alternative proposition inappropriate to assert, with respect to the use of a scalar term. The inference that the speaker may be in a certain information state is the result of the scalar implicature we get from the utterance, and it is related to a certain scope of the negation operator. One global metalinguistic negation of a strong alternative can be mapped to a set of scalar implicatures with

[9] In situation 3, the utterance in question is not quite false, but the speaker should have a good reason for not being informative sufficiently. So what scalar implicature is plausible depends on how significant the number of the students who read all. In this respect, scalar implicatures are gradient. This shows that the mechanism of deriving scalar implicatures must be flexible enough to allow for scalar implicatures of various strengths.

the descriptive negation having various scopes.

So far I have claimed that the negation operator involved in calculating scalar implicatures is metalinguistic. This idea allows more than one scalar implicature. The question is what determines the actual scalar implicature of a sentence. The first factor is **informativeness**. In (26), stronger scalar implicatures are obtained, with the negation operator having narrow scope, and they are preferred. Since preferred interpretations are readings in which the negation operator has narrower scope, it might be supposed that local scalar implicatures are preferred. However, there are cases where local scalar implicatures lead to weaker interpretations, and in such cases global scalar implicatures are preferred. This is illustrated in (18), which is repeated below.

(18) a. It's possible that Paul ate some of the eggs. (Hirschberg 1985, (75a))
b. NOT(It's possible that Paul ate all of the eggs)
c. It is possible that NOT(Paul ate all of the eggs)

By uttering the sentence, the speaker implicates 'NOTM[It's possible that Paul ate all of the eggs]'. When the speaker is not in a position to assert the stronger alternative proposition, it might be because it is possible, but not necessary, that Paul did not eat all of the eggs, as given in (18c). But if it is not necessary that Paul did not, it is also possible that Paul did. This makes (18c) trivial. A more significant inference from (18a) is that the speaker is in an information state where it is not possible that Paul ate all of the eggs, which corresponds to (18b). Then (18b) is the preferable scalar implicature. Notice that (18b) is stronger than (18c).

Consider an example in which a scalar term is within the scope of an existential quantifier. We have seen that the global scalar implicature is preferred in the following example.[10]

(14′) a. Some students who drank some of the beer were allowed to drive.
b. No students who drank all of the beer were allowed to drive. (global)
c. Some students who drank some, but not all, of the beer were allowed to drive. (local)

In this example, the speaker is not in a position to assert that some students who drank all of the beer were allowed to drive. It might be that (s)he does not have evidence for asserting the stronger proposition with *some* replaced with *all*. It means that (s)he has not seen or known of students who drank all of the beer and were allowed to drive. If the speaker knows of any such students, (s)he would have made the stronger claim, unless it is at issue whether students who drank SOME, but NOT ALL, of the beer were allowed to drive. The local weaker scalar implicature is plausible only when there is an actual or potential claim that no students who drank some of the beer were allowed to drive. Otherwise, the global scalar implicature is the intuitively correct one. So we can conclude that stronger implicatures are preferred.

4.3 Cancellation of scalar implicatures

We have seen that global scalar implicatures are preferred in some cases, and local scalar implicatures are preferred in other cases, and that interpretations preferred are stronger than less preferred ones, regardless of whether they are global or local scalar implicatures. Despite the fact that stronger scalar implicatures are preferred, we know that there are cases where

[10] The statement of possibility in (18) is also an example of an existential quantifier over possible worlds. The domains of quantification are different, but since they show the same tendency of preference, what we have observed here is systematic.

stronger ones are canceled by overt assertions or other contextual information. So we have to admit another factor that determines actual scalar implicatures. I will call it **contextual plausibility**. When stronger scalar implicatures are canceled contextually, weaker ones are checked next as candidates for actual scalar implicatures. We have seen that when a scalar term is within the scope of a universal quantifier, local negation leads to a stronger scalar implicature. So in (28), Chierchia (2004) claims, a scalar implicature arises from local negation.

(28) Every student wrote a paper or made a classroom presentation. (Chierchia 2004)

(16′) Every student at MIT has read LGB or Syntactic Structures.

In (28), when a student is supposed to do an assignment, each student is likely to satisfy the minimal requirement. They are likely to do only one of the two options. So the implicature from local negation is contextually plausible. In (16′), on the other hand, students studying linguistics at MIT are likely to read as many syntax books as possible, especially if they are written by Chomsky. So the stronger implicature that every student read only one of the two books is canceled. Then the next strongest one that not every student read both is checked to see whether it is compatible with the context.

Consider examples of obligation/requirement and permission statements. It has been pointed out that numeral terms have ′at least′ reading in obligation or requirement statements and ′at most′ readings in permission statements, as illustrated below.[11]

(29) Mary must read three books for the exam.

[11]See Carston (1998) for more detailed discussion of number terms in obligation and permission statements.

(30) Mary may take three books home.

This shows that a numeral, or more generally a scalar term, is understood as the lower limit in an obligation statement, and as the upper limit in a permission statement.[12] This applies to the following example too.

(31) Everyone must solve some of the problems.

(32) a. NOT(everyone must solve all of the problems)
b. Everyone NOT(must solve all of the problems)
c. Everyone must NOT(solve all of the problems)

In this example, there are two scope-taking expressions, so there are three possible scopes of negation. And the last one is considered in neither globalistic nor localistic analyses. In (31), the use of *some* implies that it is the lower limit, so *all* does not have to be excluded. For this reason, it is more natural to allow students to solve all the problems. (32c) is excluded. Instead, the obligation must be loosened on the number of the students as in (32a), or the obligation itself must be loosened, as in (32b). If there is someone who must solve all of the problems, (32b) is cancelled and the actual scalar implicature becomes (32a).

[12]This can be accounted for by the same account. In a simple-minded semantics of obligation/permission, an obligation statement is universal quantification over possible worlds, and if Mary reads (at least) three books for the exam in every possible world accessible with respect to deontic modality, there are possible worlds in which Mary read more. In this case it is inappropriate to assert that Mary must read four books for the exam, because there are possible worlds in which Mary read only three books. So 'three' is the minimal number. On the other hand, a permission statement is existential quantification over possible worlds. If Mary takes (at least) three books home in one of the possible worlds accessible with respect to deontic modality, 'three' must be the maximum number. If there are possible worlds in which Mary takes more than three books home, it is appropriate to say Mary may take four (or more) books home, which makes the original permission statement inappropriate.

This can be compared with permission.

(33) a. Every student is allowed to read some of the comic books.
b. Not every student is allowed to read all of the comic books.
c. Every student is not allowed to read all of the comic books.
d. Every student is allowed not to read all of the comic books.

In a permission statement, a given amount is understood to be the upper limit. So it is quite natural not to allow students to read all of the comic books. This could lead to the local scalar implicature, as given in (33d). However, this interpretation is odd because *not all* can hardly be an upper limit. Instead, the negation has scope over the operator of permission. This leads to the scalar implicature (33c), which is predicted in neither globalistic nor localistic accounts. If there are students allowed to read all of them, the implicature is further weakened to (33b).

We have discussed an example in which a scalar term is embedded in the scope of an existential quantifier. (14′) is such a case.

(14′) a. Some students who drank some of the beer were allowed to drive.
b. No students who drank all of the beer were allowed to drive. (global)
c. Some students who drank some, but not all, of the beer were allowed to drive. (local)

In this example, the global scalar implicature is the stronger one and it survives. This shows that when a scalar term is embedded in the scope of an existential quantifier, a global scalar implicature can survive. Chierchia (2004), however, discussed a similar example in which the local scalar implicature is the preferred one, as evidence for a localistic approach. If it is simply a case where contextual plausibility plays a role, it should be possible to find contexts where the global scalar implicature can survive. I will show that it is. Consider the following example.[13]

(34) a. Some students who read some of the stories will get a good grade.
b. NOTM(some students who read all of the stories will get a good grade)
c. NOT(some students who read all of the stories will get a good grade)
d. Some students who NOT(x read all of the stories) will get a good grade.

In my analysis, the negation operator is metalinguistic, as given in (34b). This can be understood as meaning (34c). The global implicature, however, is taken to be implausible: if students who read some of the stories get good grades, students who read all the stories are more likely to get good grades. So the stronger implicature is canceled. Considering the factor of contextual plausibility, this is normally what happens. However, if the cancellation is a contextual matter, we should be able to come up with a context where the stronger reading can survive.

[13]In this example, the actual implicature is calculated from 'NOTM(some students who read all of the stories will get a good grade)'. So it is possible that the negation operator could be associated with the predicate will get a good grade. This possibility, however, is excluded because it is taken to be irrelevant. The scalar implicature from the use of some can be calculated only when the negation is associated with some part containing it.

One possible context as such is one where no students read all of the stories, and such an inference is also a possible scalar implicature. When it is assumed that there are students who read all of the stories, we get the implicature (34d). This is not what is predicted by other globalistic accounts or localistic accounts.

A problem with this weaker scalar implicature is that the students mentioned in the implicature are not guaranteed to be the same group as those mentioned in the assertion.[14] This problem can be solved by introducing variable assignments in a context within the framework of dynamic semantics, but I will not give a concrete explanation.

5. Conclusion

The main claim of this paper is that the negation operator involved in calculating scalar implicatures is metalinguistic. Metalinguistic negation is used when a sentence as a whole is not appropriate to assert. Inappropriateness might come from anywhere in a sentence, and ultimately an interpretation is obtained by associating the negation with the source of the inappropriateness. This makes it possible that a globalistic analysis can give rise to the same effect as local negation. Problems with previous globalistic analyses come from the assumption that negation involved in calculating scalar implicatures is descriptive, since global negation always gives rise to global scalar implicatures. This comparison is especially

[14]A similar point is made as to presupposition projection. To solve this problem, Beaver (1992) proposed a ∂ operator and put a presupposition in the assertion so that no variable binding problem will occur. This is possible in presupposition projection because a presupposition is triggered locally. However, scalar implicatures are obtained from whole utterances, and this kind of solution does not seem possible.

clear in scalar implicatures from conjunctive and disjunctive sentences.

When a scalar term is embedded in a scope-taking operator, metalinguistic negation makes it possible to get more than one scalar implicature, depending on the scope of the negation with respect to the operators. This allows us to explain why global scalar implicatures are preferred in some cases and local scalar implicatures are in other cases, even in the same semantic structures. Among possible scalar implicatures, there is a partial ordering with respect to informativeness, and a more informative scalar implicature is preferred, if possible. One important fact is that neither global negation nor local negation always gives rise to stronger readings. The crucial thing is whether the reading is stronger or not, not whether it is obtained by global or local negation. Preference for stronger readings is what is observed in other pragmatic inferences like presupposition projection. In this respect, the generalization seems to prevail in pragmatic inferences in general.

I have discussed preference for stronger readings in terms of scopes of negation, but the structural notion of scope may not be exactly what we need. Metalinguistic negation is just an indication of inappropriateness in relation to the use of a scalar term. From the inappropriateness, we can come up with a set of information states in which the speaker can be, and the set constitutes a partial order with respect to the information they contain. The reason I discuss these in terms of the scopes of negation is that possible information states are obtained by the corresponding scopes of negation. This means that when we infer the speaker's information state from an utterance, the semantic structure of the utterance guides us to get the scalar implicature which will lead to that information state, but it does not necessarily mean that calculation of scalar implicatures is a matter of linguistic structure. This opens the possibility that some scalar implicatures can be obtained which cannot be accounted for by simply the notion of scope, but I leave it as an issue to be studied further.

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Yeom, Jae-Il
Dept. of English Language and Literature
Hongik University
72-1 Sangsu-dong, Mapo-gu, Seoul 121-791, Korea
Phone: 02) 320-1770
E-mail: jiyeom@hongik.ac.kr

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